

CO2 AT1

NAME: A K Selva Prabhu

REG NO: 192421224

COURSE CODE: CSA0904

COURSE NAME: PROGRAMMING FOR JAVA

1

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name (SELVA PRABHU A K) and registration number (192421224RA). The main content area displays Question 1: "Student Registration System" (10 marks). The question text states: "Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses. (a) Enumerate three Collection interfaces suitable for this system. (b) Design a Collection structure (diagram/representation) to store students and their enrolled courses. (c) Write a Java program using HashSet and HashMap with Generics to: • Add students • Map students to courses • Display data using Iterator". The "Your Code" section shows a Java program that uses HashSet and HashMap to store student data. The "Input" section shows "stdin input". The "Output" section shows the program's output: "Student : Priya Course : Python Student : Kumar Course : C Student : Arun Course : Java".

SIMATS ENGINEERING SIMATS Online Programming Lab
JAVA PROGRAMMING

SELVA PRABHU A K 192421224RA

QUESTIONS

- Q1 Student Registration System 10 marks
- Q2 E-Commerce Cart System 10 marks
- Q3 Library Management System 10 marks
- Q4 Employee Data Processing 10 marks
- Q5 Online Voting System 10 marks

5/5 answered

Q1 Student Registration System 10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

(a) Enumerate three Collection interfaces suitable for this system.

(b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.

(c) Write a Java program using HashSet and HashMap with Generics to:

- Add students
- Map students to courses
- Display data using Iterator

Your Code

```
1 import java.util.*;
2
3 class Student {
4
5     public static void main(String args[]) {
6
7         HashSet<String> students = new HashSet<>();
8
9         students.add("Arun");
10        students.add("Priya");
11        students.add("Kumar");
12
13        HashMap<String, String> course = new HashMap<>();
```

Input

stdin input

Output

```
Student : Priya
Course : Python
Student : Kumar
Course : C
Student : Arun
Course : Java
```

2

The screenshot shows the SIMATS Online Programming Lab interface. The header includes the SIMATS Engineering logo, the lab name, and the user's name (SELVA PRABHU A K) and registration number (192421224RA). The main content area displays Question 2: "E-Commerce Cart System" (10 marks). The question text states: "Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order. (a) List three Collection classes applicable for cart management. (b) Represent how cart items are stored using a List-based structure. (c) Write a Java program using ArrayList and Iterator to: • Add items • Remove an item • Display all items". The "Your Code" section shows a Java program that uses ArrayList to store cart items. The "Input" section shows "stdin input". The "Output" section shows the program's output: "Cart Items Laptop Keyboard".

SIMATS ENGINEERING SIMATS Online Programming Lab
JAVA PROGRAMMING

SELVA PRABHU A K 192421224RA

QUESTIONS

- Q1 Student Registration System 10 marks
- Q2 E-Commerce Cart System 10 marks
- Q3 Library Management System 10 marks
- Q4 Employee Data Processing 10 marks
- Q5 Online Voting System 10 marks

5/5 answered

Q2 E-Commerce Cart System 10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

(a) List three Collection classes applicable for cart management.

(b) Represent how cart items are stored using a List-based structure.

(c) Write a Java program using ArrayList and Iterator to:

- Add items
- Remove an item
- Display all items

Your Code

```
1 import java.util.*;
2
3 class Cart {
4
5     public static void main(String args[]) {
6
7         ArrayList<String> item = new ArrayList<>();
8
9         item.add("Laptop");
10        item.add("Mouse");
11        item.add("Keyboard");
12
13        item.remove("Mouse");
```

Input

stdin input

Output

```
Cart Items
Laptop
Keyboard
```

3



SIMATS Online Programming Lab

JAVA PROGRAMMING

 SELVA PRABHU A K

192421224RA

QUESTIONS

Q1 Student Registration System 10 marks

Q2 E-Commerce Cart System 10 marks

Q3 Library Management System 10 marks

Q4 Employee Data Processing 10 marks

Q5 Online Voting System 10 marks

5/5 answered

Q3 Library Management System 10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

(a) Name three Map implementations in Java.

(b) Design a schema-like representation using Map for storing ISBN and book details.

(c) Write a Java program using HashMap to:

- Insert book details
- Retrieve a book
- Display all books

Your Code

```

1 import java.util.*;
2
3 class Library {
4
5     public static void main(String args[]) {
6
7         HashMap<Integer, String> book = new HashMap<>();
8
9         book.put(101, "Java");
10        book.put(102, "Python");
11        book.put(103, "C");
12
13        System.out.println("Book : " + book.get(102));

```

Input

stdin input

Output

```

Book : Python
All Books
101 Java
102 Python
103 C

```

4



SIMATS Online Programming Lab

JAVA PROGRAMMING

 SELVA PRABHU A K

192421224RA

QUESTIONS

Q1 Student Registration System 10 marks

Q2 E-Commerce Cart System 10 marks

Q3 Library Management System 10 marks

Q4 Employee Data Processing 10 marks

Q5 Online Voting System 10 marks

5/5 answered

Q4 Employee Data Processing 10 marks

Suppose that a company processes employee records and filters employees based on salary.

(a) List three advantages of Generics in Collections.

(b) Represent how employee data is stored using List with Generics.

(c) Write a Java program using Iterator to:

- Store employee objects
- Display employees with salary > 50,000

Your Code

```

1 import java.util.*;
2
3 class Employee {
4
5     int id, salary;
6     String name;
7
8     Employee(int i, String n, int s) {
9         id = i;
10        name = n;
11        salary = s;
12    }
13
14    public static void main(String args[]) {

```

Input

stdin input

Output

```

Salary Above 50000
1 Arun 60000
3 Kumar 70000

```

**SIMATS**
ENGINEERING

SIMATS Online Programming Lab
JAVA PROGRAMMING

 SELVA PRABHU A K
192421224RA

QUESTIONS

Q1
Student Registration System
10 marks

Q2
E-Commerce Cart System
10 marks

Q3
Library Management System
10 marks

Q4
Employee Data Processing
10 marks

Q5
Online Voting System
10 marks

5/5 answered

Q5 Online Voting System 10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.
(a) Identify three suitable Collection types for this system.
(b) Design a structure using Set and Map for voters and vote counting.
(c) Write a Java program using HashSet and HashMap to:
• Record votes
• Prevent duplicate voting • Display results

Your Code

```
1 import java.util.*;
2
3 class Voting {
4
5     public static void main(String args[]) {
6
7         HashSet<String> voters = new HashSet<>();
8         HashMap<String, Integer> vote = new HashMap<>();
9
10        vote.put("A", 0);
11        vote.put("B", 0);
12
13        String name = "Arun";
14        String candidate = "A";
```

Input

stdin input

Output

Voting Result
A = 1
B = 0

